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Roll No

IT-8003 (1) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Digital Image Processing
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

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| 1. a) List and explain the fundamental steps in digital image processing. | 7 |
| b) Discuss briefly about the following | 7 |
| i) Neighbour of a pixel | |
| ii) Connectivity | |
| 2. a) What is Hadamard transform and specify its properties? | 7 |
| b) Explain discrete cosine transform and specify its properties. | 7 |
| 3. a) Explain about Image subtraction, averaging and image smoothing. | 7 |
| b) Discuss the Histogram based processing. | 7 |
| 4. a) Draw and explain a general compression system model. | 7 |
| b) Explain lossless and lossy predictive coding. | 7 |
| 5. a) What is meant by Image restoration and explain the degradation model. | 7 |
| b) How to detect discontinuities by point detection, line detection and edge detection model? | 7 |
| 6. a) Discuss Edge linking and Graph theoretic techniques. | 7 |
| b) Discuss briefly region oriented segmentation. | 7 |
| 7. a) Explain about morphological hit-or-miss transform. | 7 |
| b) Discuss about boundary descriptors, shape descriptors and regional descriptors. | 7 |
| 8. Write short notes on any two of the following: | |
| a) Walsh transform | 7 |
| b) Median filtering | 7 |
| c) Dilation and erosion | 7 |
